Crystalline - de Rham comparison

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(S, I, γ) base PD-scheme. $\begin{matrix} X \\ \downarrow \\ S \end{matrix}$ loc. f.t. on which γ extends.

$\Rightarrow$ $\mathrm{Crys}(X/S)$ crystalline site

$$\begin{tikzcd} U \arrow[r] \arrow[d, phantom, "\cap"] & T \arrow[dd] \\ X \arrow[d] & \\ S \arrow[r, equal] & S \end{tikzcd}$$

PD-thickening /S compatible with γ.

For $\mathcal{F}$ a crystal, $H^i_{crys}(X/S, \mathcal{F}) := R^i\Gamma(\mathcal{F})$

Th. We can compute

$$H^\bullet_{crys}(X/S, \mathcal{F}) = \mathbb{H}^\bullet(X_{Zar}, \square)$$

Last time: crystals = HPD stratification

Fix (locally) a closed embedding

$$\begin{tikzcd} X \arrow[r, hook, "i"] \arrow[dr] & Y \arrow[d] \\ & S \end{tikzcd}$$

where $Y \to S$ is smooth.

+ γ extends to Y.

$$\begin{tikzcd} & T \arrow[dr] & \\ U \arrow[ur] \arrow[dr] & \Rightarrow & P_{X,\gamma}(Y) \arrow[dr] & \\ & X \arrow[ur] \arrow[rr, "i"] & & Y \end{tikzcd}$$

PD-scheme thickening of X

final among such.

$$P_{X,\gamma}(Y) \leftleftarrows P_{X,\gamma}(Y \times_S Y) \Lleftarrow P_{X,\gamma}(Y \times_S Y \times_S Y)$$

Ex. $S = \mathrm{Spec}\, R$, $X = Y = \mathrm{Spec}\, R[x] = \mathbb{A}^1_R$

$R[x] \otimes_R R[x] \to R[x]$

$X \times_S X \overset{\Delta}{\hookleftarrow} X$

$P_{X,\gamma}(Y) = P_{X,\gamma}(X) = X$

$$P_{X,\gamma}(X \times_S X) = \mathrm{Spec}\ R[x \otimes 1]\langle 1, \overbrace{\underset{\substack{\parallel \\ x\otimes 1 - 1 \otimes x}}{dx}, (dx)^{[2]}, (dx)^{[3]}, \dots}^{\text{PD-ideal}} \rangle$$

free module

$\uparrow$ $R[x] \otimes_R R[x]$

Steps. I. Reduce to when $X = Y$ smooth $/S$

II. Identify $H_{crys}^{\bullet}$ with a Čech-Alexander complex

III. Compute its cohomology & relate to de Rham complex

I. Prop. $\mathcal{F}$ a crystal on X/S, $i_{crys,*}\mathcal{F} \in Sh(Crys(Y/S))$ is a crystal computed by: given $(U, T, \delta) \in Crys(Y/S)$, defined

$$\left(U \cap X,\ P_{I\mathcal{O}_T + I_{U \hookrightarrow T}}(T),\ \bar{\delta}\right) \in Crys(X/S)$$

$$\left(i_{crys*}\, \mathcal{F}\right)(U, T, \delta) = \mathcal{F}\left(U \cap X,\ P_{I\mathcal{O}_T + I_{U \hookrightarrow T}}(T),\ \bar{\delta}\right)$$

$\Rightarrow$ i_{crys*} is exact. $\Rightarrow$ $R\, i_{crys*} = i_{crys*}$.

$$H_{crys}^{*}(X/S, \mathcal{F}) = H_{crys}^{*}(Y/S,\ i_{crys*}\mathcal{F})$$

$$\mathcal{G} = (i_{crys*}\mathcal{F})(Y, Y, \text{triv}) = \underline{\mathcal{F}(X,\ P_{X,Y}(Y),\ \delta)}$$

think of as a quasi-coherent $\mathcal{O}_Y$-module with quasi-nilp connection

$$\left\{ \begin{matrix} \text{Crystal for} \\ Y/S \end{matrix} \right\} \longrightarrow \left\{ \begin{matrix} \text{quasi-coherent } \mathcal{O}_Y\text{-mod} \\ \text{w/ quasi-nilp inc. connection} \end{matrix} \right\}$$

$$\mathcal{G} \longmapsto \mathcal{G}(Y, Y, \text{triv})$$

Th. $H_{crys}^{\bullet}(X/S, \mathcal{F}) = H_{crys}^{\bullet}(Y/S, \mathcal{G}) = \mathbb{H}^{\bullet}(Y/S,\ \mathcal{G} \xrightarrow{\nabla} \mathcal{G} \otimes \Omega^1_{Y/S} \xrightarrow{\nabla} \mathcal{G} \otimes \Omega^2_{Y/S} \to \cdots)$

X_{Zar}

II. Instead $u_{Y/S,*} : Sh(Crys(Y/S)) \to Sh(Y_{Zar})$

Show $Ru_{Y/S,*}\, \mathcal{G} \simeq \left[\mathcal{G} \xrightarrow{\nabla} \mathcal{G} \otimes \Omega^1_{Y/S} \xrightarrow{\nabla} \cdots\right]$

localize Y so that it is affine

Covers in $Crys(Y/S)$ are Zariski covers + $\mathcal{G}$ is a crystal $\rightsquigarrow$ on affine no higher cohomology

sheaf cohomology = sheafification of derived sheaves

$$R\Gamma(\mathcal{F}) = \varprojlim_{(U,T,\delta)} R\Gamma(\mathcal{F}|_{(U,T,\delta)})$$

$$= \varprojlim \left(R\Gamma(\mathcal{F}|_{P_Y(Y)}) \rightrightarrows R\Gamma(\mathcal{F}|_{P_Y(Y \times_S Y)}) \substack{\to\\\to\\\to} R\Gamma(\mathcal{F}|_{P_Y(Y\times_S Y \times_S Y)}) \substack{\to\\\to\\\to\\\to} \cdots \right)$$

$$\overset{\text{affine}}{=} \varprojlim \left(\mathcal{F}(P_Y(Y)) \rightrightarrows \mathcal{F}(P_Y(Y\times_S Y)) \substack{\to\\\to\\\to} \cdots \right)$$

① q-coh. cohomology vanishes on affines

② $* \leftarrow Y \leftleftarrows P_Y(Y \times_S Y) \substack{\leftarrow\\\leftarrow\\\leftarrow} P_Y(Y\times_S Y \times_S Y)$ is a hypercover of $* \in Sh(Crys(Y/S))$

Def. $\mathcal{G}(Y) \xrightarrow{p_1^* - p_2^*} \mathcal{G}(P_Y(Y\times_S Y)) \xrightarrow{p_{12}^* - p_{13}^* + p_{23}^*} \mathcal{G}(P_Y(Y\times_S Y\times_S Y)) \longrightarrow \cdots$

Čech-Alexander complex

Computes $R\Gamma_{crys}(Y/S, \mathcal{G})$.

III. $\mathcal{G} = \mathcal{O}_{Y/S}$, $Y = \mathbb{A}^1_S$.

$$R[x] \xrightarrow{d} R[x]\langle 1, d, d^{[2]}, \dots \rangle \longrightarrow R[x]\left\langle (1), \begin{pmatrix} d \\ e \end{pmatrix}, \begin{pmatrix} d^{[2]} \\ de \\ e^{[2]} \end{pmatrix}, \dots \right\rangle$$

Def $F^p\mathcal{O}(P_Y(Y^n)) := \left(\text{ideal gen. by } x^{[k]},\ k\geq p,\ x\in \text{pd-ideal}\right)$

$F^p g(P_Y(Y^n)) = F^p\mathcal{O}(P_Y(Y^n)) . g(P_Y(Y^n))$

Lem. $gr^p\mathcal{O}(P_Y(Y^n)) = \Gamma^p(\Omega_{Y/S}^{\oplus n})$ where $\Gamma^p \subset (-)^{\otimes p}$ symmetric ~~tensors~~

i.e. divided symmetric powers

$$gr^p g(P_Y(Y^n)) \simeq g(Y)\otimes\Gamma^p(\Omega_{Y/S}^{\oplus n})$$

$$E_1^{p,q} = H^{p+q}(gr^p(g(P_Y(Y^\bullet)))) \Rightarrow H^{p+q}(\text{Čech- Alexander})$$

Prop. If M is a finite proj. R-mod, define

$$M_0^n = \{(x_0,\dots,x_n)\in M^{n+1} : \sum x_i = 0\}$$

Consider $\Gamma^p(M_0^0) \xrightarrow{\Gamma^p(z_0)-\Gamma^p(z_1)} \Gamma^p(M_0^1) \to \Gamma^p(M_0^2) \to \cdots$

$\Gamma^p(M_0^0) = 0$

$\Gamma^p(z_0)-\Gamma^p(z_1)+\Gamma^p(z_2)$

this has coh. concentrated in $\deg = p$, & $H^p = \Lambda^p M$.

Ex $p=0$

$$0\to R\xrightarrow{0} R\xrightarrow{1} R\xrightarrow{0} R\xrightarrow{1}$$

$p=1$

$$0\to M_0^1\xrightarrow{0} M_0^2\to M_0^3\to\cdots$$

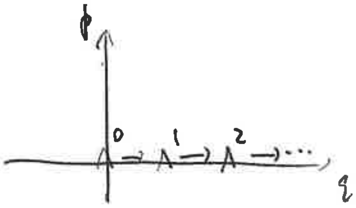
$(m,-m)\mapsto (m,-m,0)$
$\quad - (m,0,-m)$
$\quad + (0,m,-m)$
$\quad = 0$

$\Lambda^2 M \to \Gamma^2(M_0^2)/\text{im}$

$m\wedge n\mapsto (m,-m,0),(0,n,-n)$

$p=2$ $\quad 0\to\Gamma^2(M_0^1)\to\Gamma^2(M_0^2)\to\Gamma^2(M_0^3)\to\cdots$

$$E_1^{p,q} = \begin{cases} 0, & p+q \neq p \\ \Lambda^p \Omega_{Y/S}, & p+q = p \iff q = 0 \end{cases}$$



Claim. This is the de Rham complex

$$\mathcal{G}(Y) \xrightarrow{\nabla} \mathcal{G}(Y) \otimes \Omega^1_{Y/S} \xrightarrow{\nabla} \mathcal{G}(Y) \otimes \Omega^2_{Y/S} \to \cdots$$

$$\to H^*_{crys}(Y/S, \mathcal{G}) = H^*_{dR}(Y/S, \mathcal{G}(Y), \nabla)$$

$$\begin{array}{c} \parallel \\ H^*_{crys}(X/S, \mathcal{F}), \end{array} \qquad \mathcal{G} = i_{crys,*}\mathcal{F}.$$